

The Beginnings Of ASIC

Micromosaic was created by Fairchild. It was a pioneering effort that is seen as a precursor to Application Specific Integrated Circuits (ASIC), which are widely used today. Micromosaic consisted of a number of unconnected transistors. Based on the expected function to be performed, a computer program would create a map of the necessary connections which would then be used to complete the circuit. Micromosaic is also considered the first real application of CAD (Computer Aided Design).

The First Small Solid-state Calculator

Jack Kilby, John Merryman, and James VanTassel developed the first solid-state hand-held calculator at Texas Instruments. It could handle numbers with a maximum of 12 digits and perform the four basic functions. To make the calculator portable, rechargeable batteries were used, which could power the device for four hours. It would not be commercially produced until 1971.

The Floppy Disk

A team headed by David Noble from IBM created the first floppy disk. Using 8-inch magnetically-coated disks, the initial use was to load booting instructions to start IBM's mainframe computers, which till then had to be done with tape disks. Data could be stored only on one side of the disk, the storage capacity was 80 KB, and it could only be read from and not written to. Since the open disks got dirty, a cover was later added.

Laser Printing

Gary Starkweather, a Xerox employee, considered the use of lasers for printing. Starkweather started off working on creating a faster facsimile machine using a laser beam. Since the printer and facsimile machine share common areas, the leap was made. While he started to develop the idea, it was initially met with pessimism at Xerox, who could not identify any market for such a product. Eventually, the idea came to fruition at Xerox's PARC facility four years later. Xerox later sold the rights of the laser printer to HP.